

MID TERM EXAMINATION

Subject: DataBase Management System

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 40 | Date: 14-03-2026

Total Questions: 11 | Answer all questions.

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

SECTION A: MCQS

1. Which of the following is NOT typically considered a primary characteristic or benefit of a well-designed Database Management System (DBMS)? [2 Marks]
 - a) Reduced data redundancy
 - b) Enhanced data security
 - c) Increased data inconsistency
 - d) Improved data sharing

2. In a relational database, a column or a set of columns that uniquely identifies each row in a table is known as [2 Marks]
 - a) Foreign key
 - b) Candidate key
 - c) Primary key
 - d) Composite key

3. Which data model represents data in a tree-like structure, where data is organized into parent-child relationships, and a child can only have one parent? [2 Marks]
 - a) Relational model
 - b) Network model
 - c) Hierarchical model
 - d) Object-oriented model

4. The process of organizing the columns and tables of a relational database to minimize data redundancy and improve data integrity is called: [2 Marks]
 - a) Indexing
 - b) Normalization
 - c) Query optimization
 - d) Concurrency control

5. Which component of a DBMS is responsible for receiving user requests (e.g., SQL queries) and converting them into an efficient execution plan for the database engine? [2 Marks]
 - a) Storage Manager
 - b) Transaction Manager
 - c) Query Processor
 - d) Buffer Manager

6. When multiple transactions are executing concurrently, which ACID property ensures that the outcome is equivalent to some serial execution of those transactions, preventing interferences? [2 Marks]
- a) Atomicity
 - b) Consistency
 - c) Isolation
 - d) Durability

SECTION A: SHORT ANSWERS

7. Briefly explain the primary purpose of a Database Management System (DBMS). [4 Marks]
8. Differentiate between data redundancy and data inconsistency in a database system. [5 Marks]
9. List and briefly describe two significant advantages of using a DBMS over a traditional file processing system. [6 Marks]
10. What are the ACID properties in the context of database transactions? Briefly describe two of these properties. [6 Marks]
11. Explain the concept of data independence in a DBMS. Differentiate between logical and physical data independence. [7 Marks]

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